

Founding Declaration of Noetics

Fumio Miyata <https://orcid.org/0009-0008-8797-5578>

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Abstract

In the 21st century, research on intelligence has advanced rapidly under computational paradigms. Yet concepts such as computation, information processing, and statistical inference fail to capture the essential characteristics of intelligence—namely, the *generation*, *collapse*, and *phasetransitiondriven reorganization* of internal structure. Intelligence is not merely the manipulation of information; it is a **physical and geometric process of self-organization and structural reconstruction** [Fagan 2026]. This recognition has emerged independently across multiple scientific domains.

This declaration unifies these developments and formally establishes **Noetics**, a new foundational science that treats intelligence as a physical phenomenon.

Keywords: Physics of Intelligence, Noetics, Geometric Dynamics, Self-Organization, Structural Reconstruction, Substrate Invariance

1 Definition of Noetics

Noetics is the foundational scientific discipline that formulates the universal structures, laws, and dynamics of intelligence as substrateindependent first principles.

Noetics transcends the boundaries of mathematics, physics, cognitive science, and information science. It studies intelligence through three unified perspectives:

1. **Geometric Structure** Operator-level geometry based on K-theory and noncommutative geometry [Atiyah 1967; Connes and Marcolli 2006].
2. **Dissipation, Collapse, and Abstraction** Structural decay and information selection processes grounded in non-equilibrium thermodynamics and Onsager’s variational principle [Chen et al. 2024; Fagan 2026].
3. **Phase Transition, Integration, and Emergence** Dimensional jumps and the establishment of new structures based on spectral flow and the index theory of Fredholm operators [Phillips 1996; Waterstraat 2016].

These are not metaphors but **precisely defined mathematical and physical processes**.

2 Foundational Theories: PoI and PKGF

Noetics is grounded upon two core theoretical pillars.

2.1 Physics of Intelligence (PoI)

PoI defines intelligence as a **geometric dynamical process on a manifold**. The universal structure of intelligence is expressed as an irreversible threephase cycle:

- **C (Construction)**: formation of structure
- **D (Divergence)**: collapse, dissipation, abstraction
- **U (Unification)**: phasetransitiondriven integration and emergence

PoI formalizes the observation that intelligence exhibits the same structural signatures across electronic, biological, optical, and silicon substrates. This is expressed as **substrate invariance** [Fagan 2026; Ngu and Kosso 2024].

2.2 Parallel Key Geometric Flow (PKGF)

PKGF provides the **mathematical infrastructure** underlying PoI. It is a unified theory of linear operator flows that integrates:

- conservative flows (Liealgebraic structure)
- dissipative flows (elliptic operators, parabolic PDEs)
- unified flows (complex linear combination)
- spectral flow (topological invariants) [Booss-Bavnbek et al. 2005]

PKGF offers the **analytical infrastructure**—semigroup theory, elliptic theory, and spectral flow theory—required to treat these components coherently within a single Hilbertspace framework. In particular, semigroup theory based on the Hille–Yosida theorem ensures the well-posedness of the evolution [Brezis 2011].

PoI provides the *physical laws* of intelligence. PKGF provides the *mathematical realization* of those laws. Together, they constitute the foundation of Noetics.

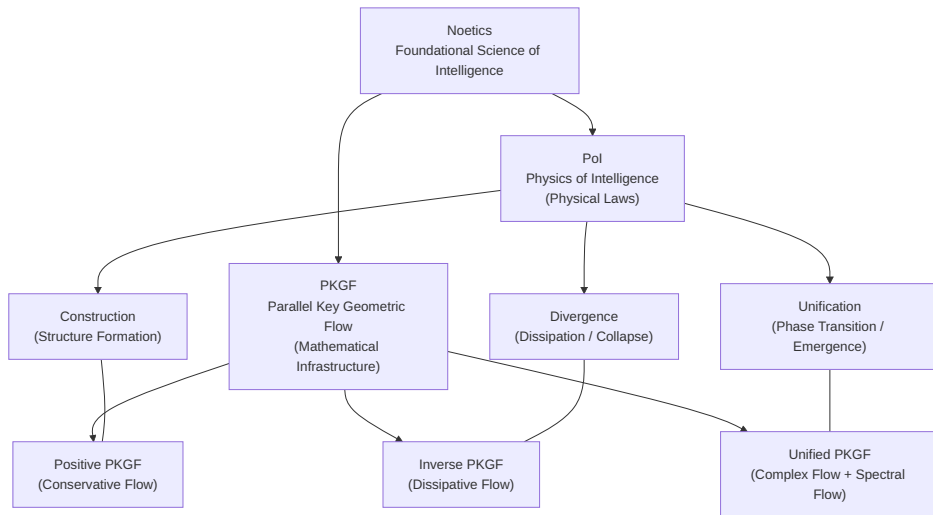


Figure 1: Diagram 1

3 Scientific Mission of Noetics

Noetics advances three primary missions.

3.1 Establishing First Principles of Intelligence

To liberate intelligence from computational metaphors and reconstruct it as a **fundamental science with the rigor of physics** [Fagan 2026].

3.2 Discovering SubstrateInvariant Universal Laws

To describe the structures shared by biological, artificial, and physical intelligent systems using:

- differential geometry
- operator theory
- topological invariants

4 Methodological Framework of Noetics

Noetics is built upon a threelayer methodological architecture:

1. **Structural Layer** Manifolds, bundles, operators, gauge groups [Melrose 1993; Tong 2018].
2. **Analytical Layer** Semigroup theory, elliptic operator theory, spectral flow [Davies 2006; Cheng 2026].
3. **Physical Layer** CDU cycle, phase transitions, substrate invariance [Fagan 2026; Ngu and Kosso 2024].

This layered structure ensures the coexistence of **mathematical rigor** and **physical interpretability**.

5 Declaration of the Founding of Noetics

Based on the above, I hereby declare:

> **Noetics is formally established as a new foundational science that treats intelligence as a physical phenomenon.** > **PoI and PKGF are designated as its foundational theoretical frameworks and will serve as the core of future academic development.**

Noetics aims to transcend the fragmentation of current intelligence research and to unify mathematics, physics, cognitive science, and engineering into a **new scientific framework for the 21st century**.

6 Conclusion

The establishment of Noetics is not merely the proposal of a new theory, but the establishment of a new scientific worldview for understanding intelligence as a universal physical phenomenon.

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